

SEINAJOKI PELMAA, Finland

WMO: 028330

Lat: 62.9381N

Lon: 22.4889E

Elev: 32

StdP: 100.94

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.8	-21.0	-27.6	0.3	-24.6	-23.4	0.5	-20.9	9.3	-2.5	8.5	-1.1	0.6	100	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	27.2	18.7	25.1	17.6	23.2	16.8	20.2	24.7	18.9	23.3	17.8	21.9	2.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.6	22.2	17.3	12.4	20.7	16.0	11.4	19.6	58.1	25.3	53.8	23.5	50.2	21.8	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.8	6.8	6.0	-26.4	29.2	5.2	2.9	-30.1	31.2	-33.2	32.9	-36.1	34.6	-39.8	36.7		
(5)			-25.6	21.5	6.0	2.2	-29.9	23.0	-33.4	24.3	-36.7	25.5	-41.1	27.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.6	-7.2	-5.9	-2.4	3.5	9.7	13.5	16.7	14.9	10.5	4.3	0.5	-3.8	(6)
(7)		DBStd	9.30	6.57	6.85	4.98	3.41	4.40	3.41	3.32	3.08	3.40	3.59	4.44	6.07	(7)
(8)		HDD10.0	2553	534	445	384	197	58	7	1	2	35	178	284	427	(8)
(9)		HDD18.3	5053	793	678	643	446	269	149	73	113	234	434	534	686	(9)
(10)		CDD10.0	578	0	0	0	2	49	112	208	154	51	2	0	0	(10)
(11)		CDD18.3	37	0	0	0	0	2	6	22	8	0	0	0	0	(11)
(12)		CDH23.3	401	0	0	0	0	41	73	226	62	0	0	0	0	(12)
(13)		CDH26.7	76	0	0	0	0	3	14	49	10	0	0	0	0	(13)
(14)	Wind	WSAvg	2.5	2.5	2.7	2.8	2.5	2.6	2.3	1.9	2.1	2.2	2.7	2.8	2.8	(14)
(15)	Precipitation	PrecAvg	565	35	25	26	29	44	58	79	71	55	55	48	41	(15)
(16)		PrecMax	707	77	55	54	76	86	113	130	161	130	117	101	71	(16)
(17)		PrecMin	447	9	1	5	4	8	16	18	18	23	10	8	14	(17)
(18)		PrecStd	78	17	14	12	17	23	24	33	35	31	29	21	18	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.2	5.4	10.0	19.5	26.9	28.9	30.4	28.2	21.3	14.4	10.5	6.8	(19)
(20)			MCWB	3.3	3.4	5.3	11.1	16.0	18.6	20.7	20.4	16.0	12.1	9.3	6.2	(20)
(21)		2%	DB	2.7	4.0	6.9	15.2	24.3	25.2	27.9	24.8	19.4	12.2	8.5	4.7	(21)
(22)			MCWB	1.8	2.5	3.9	9.3	14.8	17.2	19.8	18.0	14.9	10.6	7.6	3.6	(22)
(23)		5%	DB	1.7	2.6	5.0	12.3	21.6	22.9	25.9	22.8	17.7	11.1	7.1	3.4	(23)
(24)			MCWB	1.1	1.8	2.7	7.4	13.9	15.7	18.6	16.8	13.8	9.7	6.4	2.6	(24)
(25)		10%	DB	0.8	1.7	3.6	9.9	18.6	20.6	23.7	20.8	16.1	9.7	5.9	2.4	(25)
(26)			MCWB	0.4	1.1	1.8	6.1	12.3	15.0	17.9	15.9	13.3	8.4	5.5	1.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	3.8	5.9	11.9	17.2	20.0	22.6	21.5	16.8	12.8	9.4	6.2	(27)
(28)			MCDWB	4.1	5.2	8.4	18.0	23.7	25.3	28.3	26.3	20.1	14.0	10.3	6.6	(28)
(29)		2%	WB	2.1	2.6	4.1	9.7	16.0	18.4	20.9	19.0	15.6	11.2	7.7	3.9	(29)
(30)			MCDWB	2.5	3.6	6.4	14.7	22.7	23.7	25.5	22.8	18.4	12.0	8.4	4.5	(30)
(31)		5%	WB	1.2	2.0	3.1	7.7	14.4	16.8	19.7	17.8	14.6	10.0	6.4	2.7	(31)
(32)			MCDWB	1.6	2.7	4.7	11.4	20.9	21.5	24.0	21.4	16.8	10.9	7.0	3.2	(32)
(33)		10%	WB	0.4	1.2	2.0	6.4	12.7	15.6	18.6	16.7	13.7	8.6	5.5	1.9	(33)
(34)			MCDWB	0.9	1.7	3.4	9.7	18.2	19.8	22.9	19.7	15.7	9.6	6.0	2.4	(34)
(35)	Mean Daily Temperature Range	MDBR	6.1	6.5	8.1	9.0	11.2	10.2	10.3	9.9	8.7	6.2	3.8	4.9	(35)	
(36)		5% DB	MCDBR	4.4	4.9	8.0	14.3	15.9	14.6	13.9	12.7	10.9	7.7	4.8	4.5	(36)
(37)			MCWBR	4.1	4.1	5.4	8.9	7.9	7.6	6.7	6.6	6.1	4.4	4.2	(37)	
(38)			MCDBR	4.2	4.5	7.2	12.9	15.0	12.3	12.1	10.7	9.4	6.9	4.7	4.6	(38)
(39)		5% WB	MCWBR	4.1	3.9	5.3	8.4	7.8	6.6	6.4	6.0	6.1	6.0	4.4	4.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.239	0.295	0.320	0.359	0.355	0.353	0.375	0.361	0.336	0.318	0.258	0.352	(40)	
(41)		taud	2.058	2.185	2.276	2.333	2.424	2.450	2.411	2.444	2.504	2.381	2.137	2.570	(41)	
(42)		Ebn at Noon	394	636	778	822	862	869	835	816	769	619	365	222	(42)	
(43)		Edh at Noon	32	63	85	100	100	99	101	89	70	53	29	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.17	0.74	2.24	3.71	5.00	5.44	5.07	3.88	2.28	0.89	0.22	0.08	(44)	
(45)		RadStd	0.02	0.16	0.23	0.40	0.53	0.47	0.49	0.40	0.25	0.18	0.04	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (45 neighbors)	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page